



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO VENKATESWARA AUTO SERVICE

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

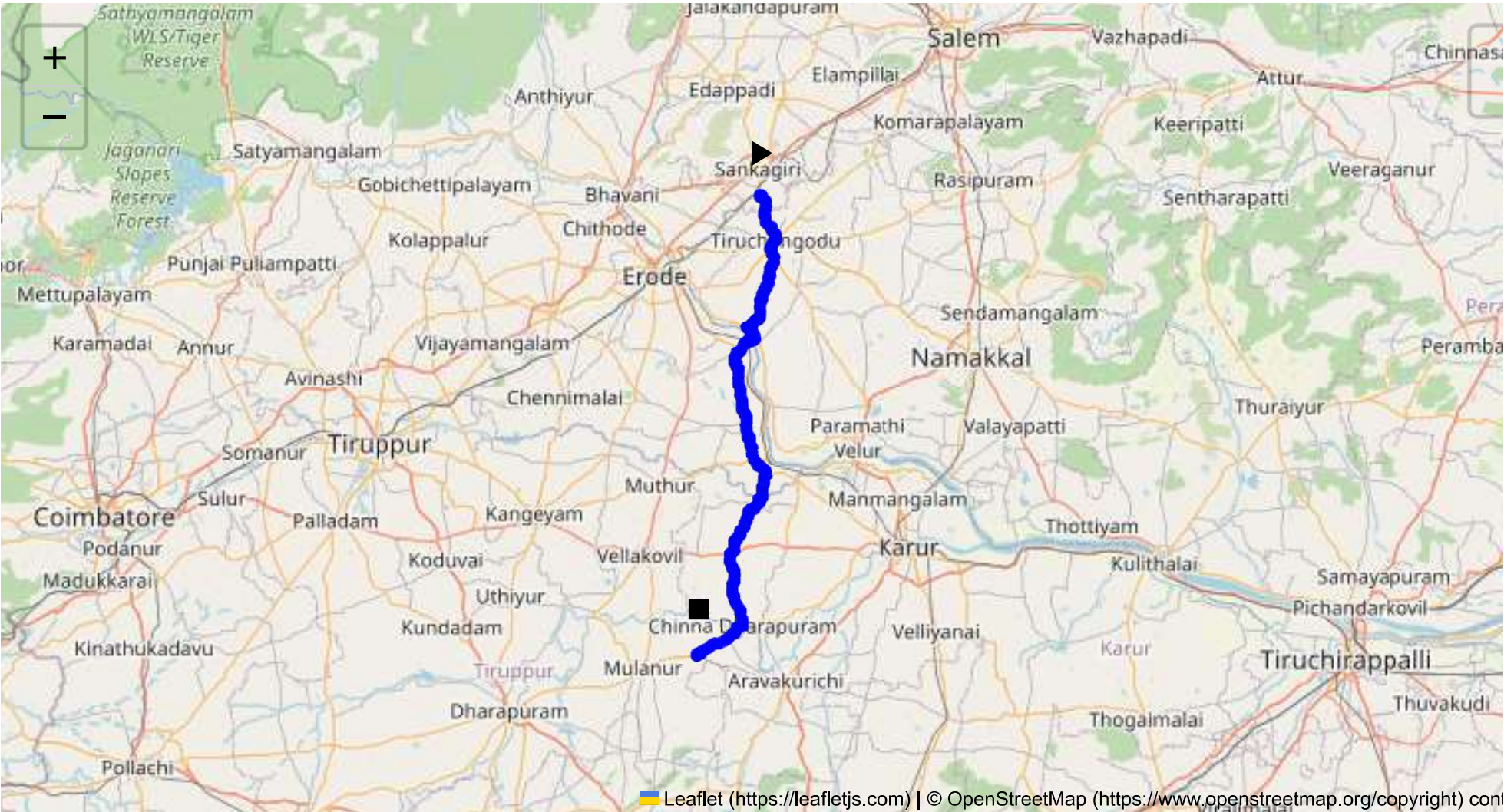
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 83.51 km
Estimated Duration: 1.9 hours
Adjusted Duration (Heavy Vehicle): 2.4 hours
Start: (11.4381, 77.8734)
End: (10.813249, 77.785988)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

Route Overview: The route from Sangagiri to Kannaivadi spans approximately 83.51 kilometers. This route traverses through the state of Tamil Nadu, utilizing a combination of state highways and rural roads. The journey typically takes about 1.92 hours by heavy vehicles carrying hazardous materials.

1. Route Map Overview: The journey primarily involves rural and semi-urban settings, starting in Sangagiri and passing through towns such as Tiruchengode and Pallipalayam before reaching Kannaivadi. The final destination is situated in a less populated region, with significant stretches of road bordered by agricultural land.

2. Typical Weather Conditions and Potential Hazards: Tamil Nadu experiences a tropical climate with distinct wet and dry seasons. The route may encounter:

- **Monsoon (June-September):** Heavy rains can lead to water-logged roads and reduced visibility. Service areas are essential during this period.
- **Summer (March-May):** High temperatures can cause road surface deterioration and heat-related stress on vehicles. Preparation for sudden weather changes is recommended.

3. Traffic Patterns and Congestion:

- **Peak Hours:** Major towns like Tiruchengode experience congestion during morning (8-10 AM) and evening (5-7 PM) peak hours.
- **Congestion-Prone Areas:** Market areas and school zones within these towns are particularly busy; plan transit times accordingly.

4. Assessment of Road Quality and Infrastructure: Major segments of the route are well-paved, primarily state highways, but some sections may have potholes or uneven surfaces, particularly on rural roads. Certain areas lack adequate lighting, increasing risk during nighttime travel.

5. Alternative Emergency Routes:

- **Via Erode:** An alternative for certain sections is to divert through Erode, utilizing NH544; however, this increases the travel distance slightly but offers better infrastructure and emergency services.

6. Local Regulations Affecting Hazardous Material Transport:

- Adhere to the Motor Vehicles Act of India concerning the transportation of hazardous materials.
- Specific permits may be required, especially through populated towns. Compliance with speed limits and hazards warning signage is mandatory.

7. Historical Incidents: Historical data indicates several minor incidents involving heavy vehicles due to sharp turns and inadequate signage. There have been some reports of overturned lorries during the monsoon due to slippery conditions.

8. Environmental Considerations and Sensitive Areas:

- **Agricultural Land:** Avoid chemical spills which can contaminate crops.
- **Water Bodies:** Particular care needed near dam or canal systems to prevent water contamination.

9. Communication Coverage: Most of the area is well-covered by telecom networks, though some rural sections may experience weaker signals. Ensure communication devices are checked and alternative contact methods are established.

10. Emergency Response Times:

- **Urban Areas (Sangagiri, Tiruchengode):** Emergency services generally respond within 20-30 minutes.

- **Rural Sections:** Response times can exceed 45 minutes. Keep emergency contact information readily available.

11. Overall Summary of Risk Assessment: The route presents moderate risks primarily related to weather and road conditions. Preparation for monsoon impacts and adherence to operating permits are key to mitigating risks. Emergency services are accessible but can be delayed in rural areas, so drivers should be prepared with additional emergency plans. Overall, by practicing careful driving and route planning, these risks can be substantially minimized.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44022, 77.87546	15 KM/Hr
3	Turn	Medium	11.39976, 77.88382	30 KM/Hr
4	Turn	Medium	11.39690, 77.88852	30 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr
6	Turn	Medium	11.37857, 77.89473	30 KM/Hr
7	Turn	High	11.33268, 77.88827	15 KM/Hr
8	Turn	Medium	11.33165, 77.88697	30 KM/Hr
9	Turn	High	11.32130, 77.88484	15 KM/Hr
10	Turn	Medium	11.32102, 77.88435	30 KM/Hr
11	Turn	Medium	11.31183, 77.88094	30 KM/Hr
12	Turn	Medium	11.31151, 77.88008	30 KM/Hr
13	Turn	Medium	11.30805, 77.87934	30 KM/Hr
14	Turn	Medium	11.30625, 77.87776	30 KM/Hr
15	Turn	High	11.29763, 77.87710	15 KM/Hr
16	Turn	High	11.29668, 77.87515	15 KM/Hr
17	Turn	Medium	11.29620, 77.87503	30 KM/Hr
18	Turn	High	11.29585, 77.87457	15 KM/Hr
19	Turn	High	11.29549, 77.87464	15 KM/Hr
20	Turn	Medium	11.29417, 77.87309	30 KM/Hr
21	Turn	Medium	11.27096, 77.87034	30 KM/Hr
22	Turn	Medium	11.27057, 77.86827	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
23	Turn	High	11.26028, 77.85633	15 KM/Hr
24	Turn	High	11.25791, 77.85522	15 KM/Hr
25	Turn	Medium	11.25287, 77.86220	30 KM/Hr
26	Turn	Medium	11.24410, 77.86671	30 KM/Hr
27	Turn	High	11.24365, 77.86668	15 KM/Hr
28	Turn	High	11.24100, 77.86030	15 KM/Hr
29	Turn	Medium	11.24196, 77.85902	30 KM/Hr
30	Turn	Medium	11.24193, 77.85850	30 KM/Hr
31	Turn	Medium	11.22478, 77.84241	30 KM/Hr
32	Turn	High	11.22377, 77.83980	15 KM/Hr
33	Turn	Medium	11.14595, 77.84958	30 KM/Hr
34	Turn	Medium	11.12977, 77.85502	30 KM/Hr
35	Turn	Medium	11.12714, 77.85403	30 KM/Hr
36	Turn	Medium	11.05412, 77.87969	30 KM/Hr
37	Turn	Medium	11.04059, 77.87686	30 KM/Hr
38	Turn	Medium	11.02806, 77.87663	30 KM/Hr
39	Turn	Medium	11.02541, 77.87459	30 KM/Hr
40	Turn	High	11.02422, 77.87229	15 KM/Hr
41	Turn	Medium	11.02248, 77.87260	30 KM/Hr
42	Turn	Medium	11.02213, 77.87236	30 KM/Hr
43	Turn	Medium	11.02054, 77.86997	30 KM/Hr
44	Turn	High	11.01822, 77.86979	15 KM/Hr
45	Turn	Medium	11.00834, 77.85882	30 KM/Hr
46	Turn	Medium	11.00475, 77.85797	30 KM/Hr
47	Turn	Medium	11.00360, 77.85683	30 KM/Hr
48	Turn	Medium	10.99820, 77.85581	30 KM/Hr
49	Turn	Medium	10.96483, 77.83744	30 KM/Hr
50	Turn	Medium	10.95683, 77.83813	30 KM/Hr
51	Turn	Medium	10.95401, 77.83691	30 KM/Hr
52	Turn	Medium	10.89518, 77.83562	30 KM/Hr
53	Blind Spot	Blind Spot	10.85240, 77.85000	10 KM/Hr
54	Turn	High	10.81414, 77.79125	15 KM/Hr
55	Turn	Medium	10.81215, 77.78718	30 KM/Hr
56	Turn	High	10.81177, 77.78603	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
6	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
10	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	clinic	Jai Sri Amman Clinic	11.1102489, 77.8579522	30 km/h	Medium
15	hospital	Government Hospital, Chinadarapuram	10.8594445, 77.8485824	30 km/h	Medium
16	hospital	Prakash Hospital	10.85157, 77.846321	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44022, 77.87546



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39976, 77.88382



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39690, 77.88852



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37868, 77.89498



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.37857, 77.89473



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.33268, 77.88827



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.33165, 77.88697



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.32130, 77.88484



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.32102, 77.88435



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.31183, 77.88094



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.31151, 77.88008



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.30805, 77.87934



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.30625, 77.87776



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.29763, 77.87710



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.27096, 77.87034



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.27057, 77.86827



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.26028, 77.85633



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.25791, 77.85522



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.25287, 77.86220



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.24410, 77.86671



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.24365, 77.86668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.24100, 77.86030



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.24196, 77.85902



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.24193, 77.85850



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.22478, 77.84241



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.22377, 77.83980



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.14595, 77.84958



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.12977, 77.85502



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.12714, 77.85403



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.05412, 77.87969



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.04059, 77.87686



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.02806, 77.87663



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.00834, 77.85882



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.00475, 77.85797



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.00360, 77.85683



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.99820, 77.85581



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.96483, 77.83744



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.95683, 77.83813



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.95401, 77.83691



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.89518, 77.83562



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.85240, 77.85000



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.81414, 77.79125



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.81215, 77.78718



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.81177, 77.78603

